Qualitative Anatomical Analysis of Unknown Bone within *Crocidura* Shoulder Girdle

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BACKGROUND

Shrews are a highly diverse set of **small, insectivorous mammals** found throughout the world. While some species are specialized for **tunneling**, others excel at **climbing, swimming, or rapid surface movement**. Despite their ecological diversity, certain aspects of their skeletal anatomy remain **insufficiently studied**.

An **undescribed bone segment** located between the **clavicle and scapula** in the *Crocidura* genus of shrews showcases this. It's function, if any, is unknown. We use **Diffusible Iodine-based Contrast-enhanced Computed Tomography (DiceCT)** to conduct a **qualitative anatomical analysis** of the shoulder girdle. By manually segmenting bones, muscles, and ligaments, we aim to identify this unknown structure and explore its **functional and evolutionary significance**.



Figure 1: Top-down view of prepared *Sorex tibetanus* Shrew

METHODS

Imaging & Software: High-resolution DiceCT scans of a *Crocidura brunnea* were processed using VGSTUDIO for 2D and 3D segmentation. Bones and soft tissues were visually identified based on density contrast in CT images.

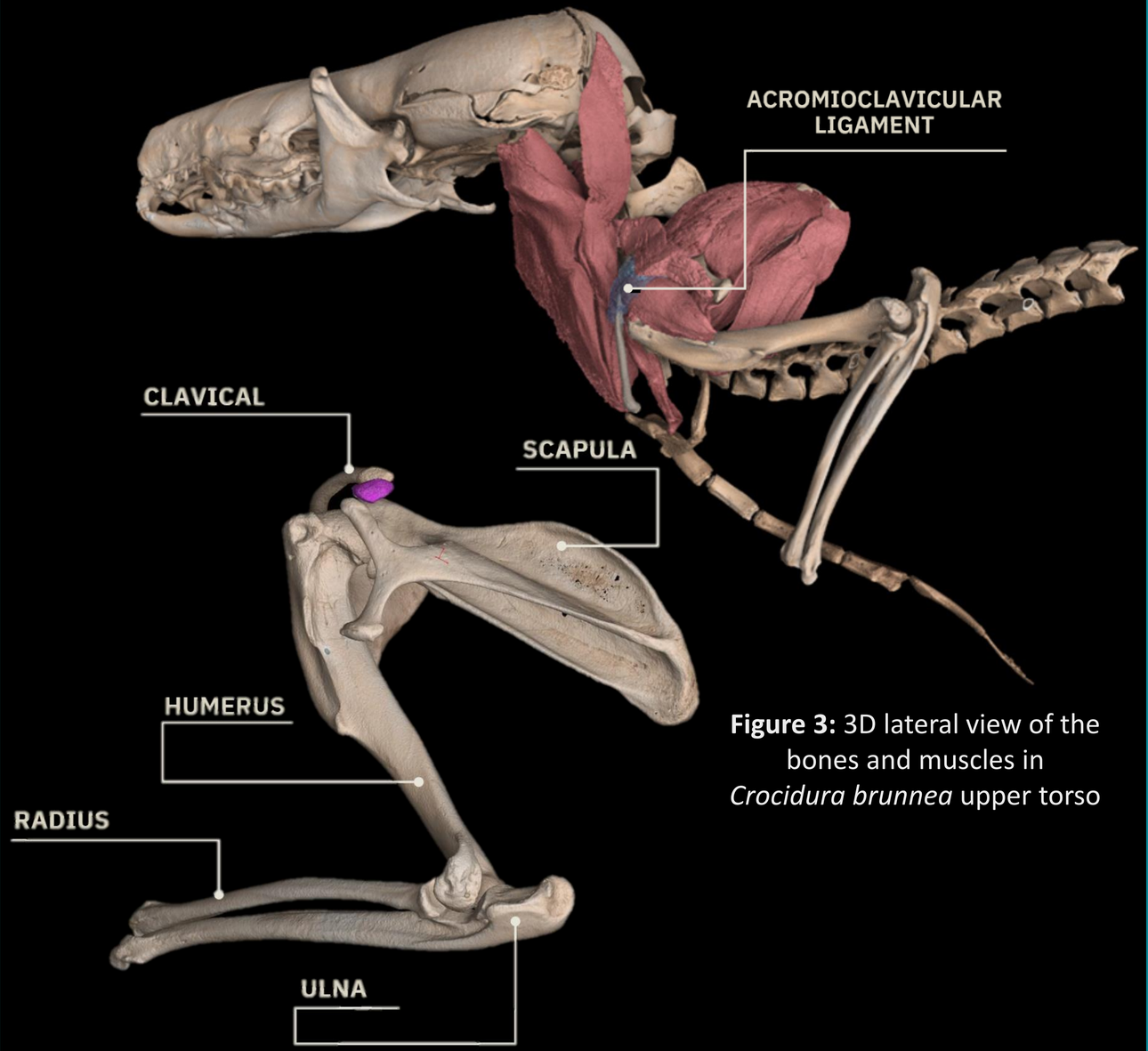


Figure 2: 3D lateral view of the bones in the Crocidura fore limb

Figure 3: 3D lateral view of the bones and muscles in *Crocidura brunnea* upper torso

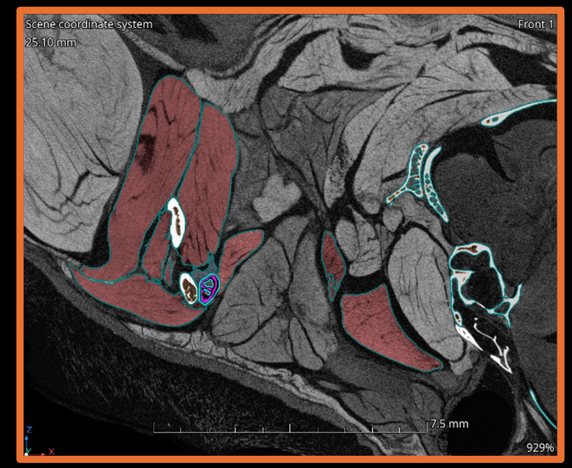


Figure 4: 2D visualization of segmented muscles and bones in *Crocidura* DiceCT

DISCUSSION

The DiceCT analysis of *Crocidura brunnea* shoulder girdles of the **undocumented bone segment** between the **clavicle and scapula** revealed that it:

- Lacks **unique muscle attachments** and **independent movement**, suggesting no clear function.
- A similar bone present in **marsupial moles** and a few **burrowing rodents** but absent in **true moles** and **Sorex shrews**.
- Other **various small undescribed bones** that may have **developed in a similar manner**.

These findings suggests it may possibly be a **sesamoid or vestigial trait**. Further studies are needed to determine its evolutionary significance and development process.

ACKNOWLEDGMENTS

Edward Stanley, Peishu Li, Ph.D., The Florida Museum Undergraduate Internship Program Committee, Florida Museum of Natural History

REFERENCES

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